LLM 3-oy 2-Dars Report

SFT — bu robot xolatida bo'lgan faqat keyingi so'zni predict qiladi modeldan xuddi odamga o'xshab javob beradigan xolatga o'tkazish.

Buni o'z bosqichlari bor buni biz katta emas kichik doirada training qilamiz Question Response ko'rinishidagi datalar b-n modelni training qilib olamiz

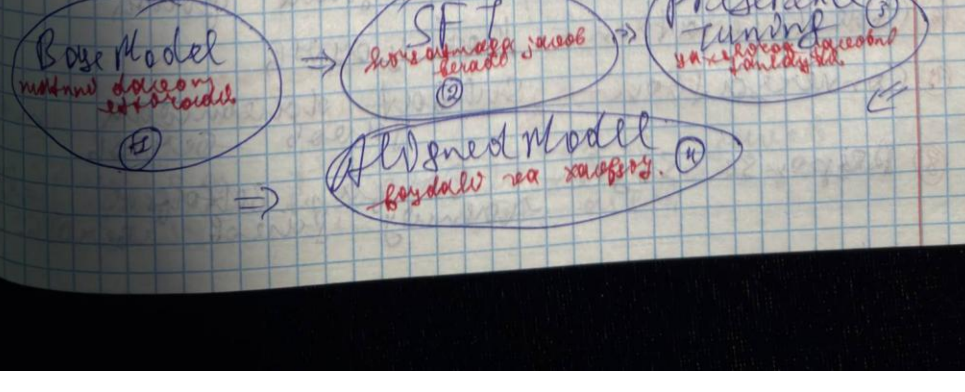
- Pre training modelni o'zimizni ko'rsatmalarimiz b-n robot xolatidan Aqlli yordamchi xolatiga o'tkazish

Post Trainingda 2 ta asosiy bo'lim mavjud SFT + Alignment

SFT — nima javob berishni o'rganadi

Alignment — Alignment yomon narsalarga javob berishni oldini oladi

SFT da instruction-response juftlik b-n o'rgatiladi. Misol u-n 2000 ta juftlik.



SFT da loss ni anglashda

Asosan javobni losini anglashga xarakat qiladi. SFT da cross-entropy loss ishlatiladi. Asosiy urg'uni javob qanchalik yaxshi chiqdi degan savolga qaratadi.

SFT quyidagilarni yaxshi o'rgatadi

Format: ro'yhat, JSON, kod bloki

Rollar: user/ assistant

Qachon to'xtash: (EOS Token)

Ko'rsatmaga amal qilish, uslub, ohang

SFT orqali biz quyidagilarga erishamiz

① Instruction Following – user ko'rsatmalarig amal qilish

② Response Formatting – javob/ni kerakli formatda yozish

③ Question Answering – savol/ga mos javob berish

④ Summarization – Matnni qisqartirib, asosiy mazmunini berish

⑤ Translation – Bir tildan boshqa tilga tarjima qilish

⑥ Code Generation – Kod yozish

⑦ Conversation – Suhbat davomida mos javob berish

⑧ Response Style – Qisqa, batafsil yoki rasmiy javob berish

SFT da Asosiy Maqsad va Strategy

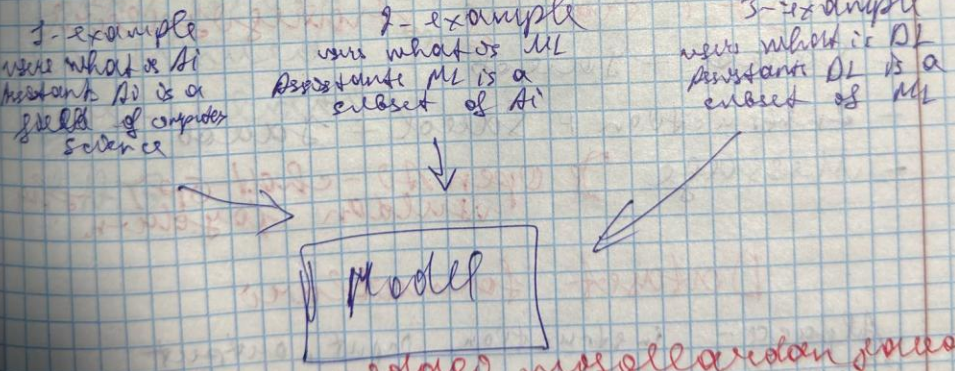
- Instruction Following qobilyatini rivojlantirish
- Instruction-Response juftliklari
- Instruction-tuned model

SFT ishlashi

Training Data => SFT Tokenization => Prediction => Loss Calculation => Backpropagation => Parameter Update

Behavior Cloning -> Bu modelga kerakli xatti-xarakatni namoyish etuvchi misollarni ko'rsatish va undan shu xatti harakatni takrorlashni o'rgatish usuli

Behavior Cloning oddiy ko'rinishi



1-example
user what is Ai
Assistant Ai is a field of computer science

2-example
user what is ML
Assistant ML is a subset of Ai

3-example
user what is DL
Assistant DL is a subset of ML

Model yuqoridagi misollardan javob berish uslubini o'rganadi

- Savolga to'g'ridan-to'g'ri javob berish
- qisqa va tushunarli jumlalar yozish
- Keraksiz matn qo'shmaslik

Loss Masking – faqat javobga e'tibor berish va loss larni faqat javoblar uchun xisoblash jarayon

Teacher Forcing – Treyning davomida modelga keyingi tokenni bashorat qilish uchun o'zining oldingi bashoratidan emas, balki haqiqiy oldingi tokenlardan foydalanish imkonini beradigan usul. (Ustoz xatoni to'g'rilab turadi doimiy)

Afzalligi: training tezlashadi, yuzlab marta tezroq training, GPU da parallel ishlash.

Kamchiliklari: Inferenceda xatoga moyil ekan, generatsiya tushunchasi zaif

SFT training strategiyalari

- Savol + Javob
- Instruction + Savol + Javob
- Message } OpenAI - chatGPT - shu usuldan foyda-n

Dataset formatlari

Alpaca – instruction, input, output
Share GPT – conversation: [{from, value}]
Message – messages [{role, content}]
prompt-completion – prompt, completion

Malumot Qayerdan Keladi

Inson yozgan => Dolly - 15k, OASST, No Robots
Synthetic (Self-Instruct) => Stanford Alpaca (52k)
Distillation (kuchli modeldan) -> Open Hermes, Tulu3 mix
Mavjud tasklardan => FLAN
(template)

Bizning Tanlovimiz

Base model: Qwen 2.5 - 0.5B
Dataset: yahma/alpaca-cleaned
Hajm: 2000 train + 200 test
Format: Alpaca => prompt - completion
Eval: Uy vazifasidagi 5 ta savol
O'zbekcha: O'zingiz yozgan 20 ta namuna

SFT va uning Alternativalari

Prompt engineering / few shot
Full SFT
LoRA / QLoRA SFT
Continued pre training
Distillation SFT
RAG